

A close-up photograph of a laboratory setting. A person wearing blue nitrile gloves is holding a clear glass petri dish filled with small, light-brown, oval-shaped seeds. The person's hands are positioned over a clipboard with a document. In the background, there are some green leafy vegetables and a black cap. The overall scene suggests a food safety or quality control laboratory.

FIGHTING FOOD FRAUD

Bruker FT-NIR for Targeted/Non-Targeted Adulterant Screening

This application note shows how hierarchical spectral analysis using Bruker FT-NIR spectroscopy and OPUS MultiEvaluation software simplifies food quality control. The technique allows for accurate assessment of both raw and final products as well as quick detection of adulteration by combining identification, conformance testing, and quantification into a single automated workflow. Its preparation-free, non-destructive method offers strong protection against food fraud in addition to operational efficiency.

Keywords or phrases:

Food fraud detection, FT-NIR spectroscopy, adulterant screening, quality control, non-targeted analysis, Bruker OPUS MultiEvaluation

INTRODUCTION

Maintaining brand reputation and product quality has led to increased focus on qualifying raw materials and ingredients used in food production. In addition to traditional quality parameters like moisture, fat, and protein, detection of adulterants plays a critical role in ensuring food safety and quality. The USP (US Pharmacopeial Convention) Food Fraud Database currently lists hundreds of incidents of economically motivated adulteration (EMA), substitution, counterfeiting or mislabelling of food products, such as olive oil and milk powder and some prominent adulterants e.g. melamine.

FT-NIR offers a valuable tool for screening almost any raw material with an excellent cost-benefit ratio and unrivalled ease-of-use. The high information content of NIR spectra provides a fingerprint of the complete sample.

Comparing the spectra of the incoming raw materials with

those measured using samples of known quality permits a non-targeted screening of adulterants or contaminants within the detection limits. If a material is tested positive on adulteration by FT-NIR, further investigations with complementary analytical methods can be carried out to determine the exact identity of the adulterant.

Method

Bruker FT-NIR spectrometers offer a fast and effective tool for quality control of raw materials, intermediate products and final products. As a non-destructive method without any sample preparation, it is already extensively used in the food industry for analysing main constituents such as protein, moisture, fat, lactose, ash, and fiber.

With the measurement of a single spectrum the sample can be evaluated in a three-step process. This enables verification of the sample identity and targeted screening for known contaminants as well as non-targeted screening for adulteration with unknown materials, followed by traditional quantification of the main constituents:

1. Identification

Identification of a sample is carried out to determine if the spectrum of an incoming raw material fits within the statistical population of authentic and previously accepted batches. With this first step it can be checked if the correct raw material was delivered and properly labelled to avoid usage of the wrong ingredient in production.

2. Conformity

In the next step the sample is further qualified using conformity test, which is a more specific evaluation of the spectrum. Each data point of the NIR spectrum is subject to a dedicated test with an individual threshold.

3. Quantification

During the quantification of the different constituents an outlier test based on the Mahalanobis Distance is performed. Again, the analysis spectrum is compared to the sample population in the individual quantification models.

CONCLUSION

Bruker Optics comprehensive spectroscopy software OPUS provides powerful tools for identification, qualification by conformity test and quantification of raw materials and finished products. The powerful MultiEvaluation package combines these individual evaluation tools to a single routine.

This allows the hierarchical evaluation of spectra using different methods, like identification of a product in a list of hundreds of products, followed by a dedicated conformity test and/or quantification of the identified product.

The software user interface links measurement and evaluation methods to a single raw material or product.

The operator only needs to select the product to be analyzed and the correct measurement and evaluation methods are performed automatically. Green checkmarks are used for passing evaluations, whereas a red cross indicates the sample has failed the test.



Figure 1. Example scheme of a MultiEvaluation method with hierarchical identification followed by conformity testing or quantitative evaluations.

Figure 2. 3D scores plots showing different material groups in an identification library.

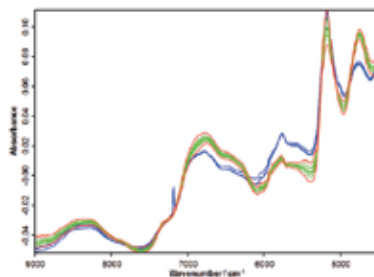
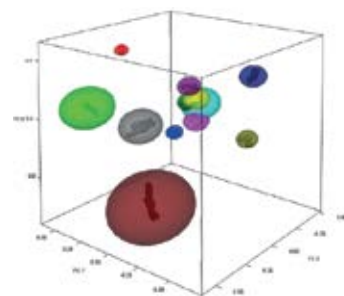


Figure 3. Display of reference spectra in a conformity test method (green) with spectra detected as outliers (blue).

Figure 4. Result screen for a quantification showing a Mahalanobis outlier for the fat evaluation.



TANGO FT-NIR Spectrometer

Touch-screen operated FT-NIR analyser for routine use in the lab and at-line.

